IGCSE 0607 Practice Questions — E6.2 — Combined Questions & Markscheme

Area and perimeter

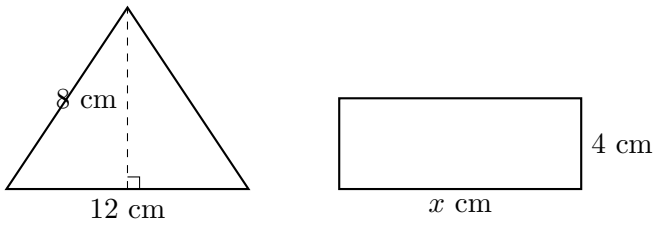
Total Questions: 9 — **Total Marks:** 60 Marks

Table of Contents & Quick Navigation

- **E6.2** Area and perimeter
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

---

1 An isosceles triangle and a rectangle are shown below.
The area of the triangle is equal to the area of the rectangle.



NOT TO SCALE

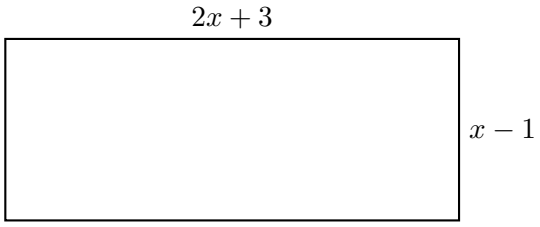
(a) Calculate the value of x.

$x =$ ..... **[2]**

(b) Calculate the perimeter of the isosceles triangle.

..... cm **[3]**

2 The diagram shows a rectangle.
All measurements are in centimetres.



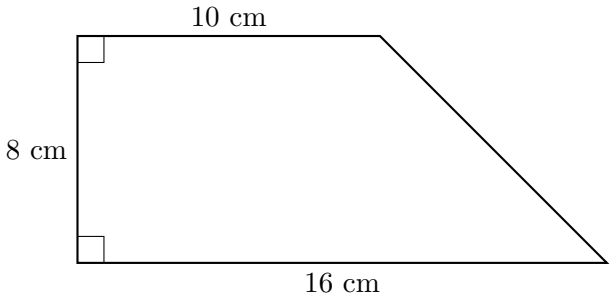
(a) The perimeter of the rectangle is 34 cm.
Find the value of x.

$x =$ ..... **[3]**

(b) Calculate the area of the rectangle.

..... cm^2 **[2]**

3 The diagram shows a right-angled trapezium.



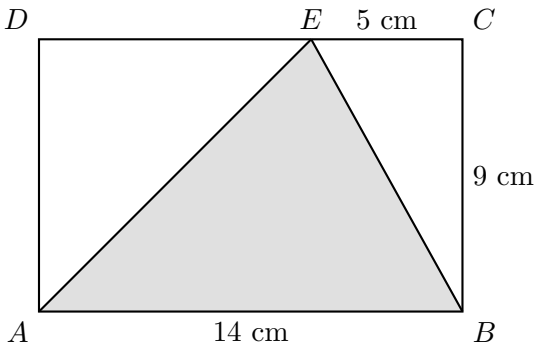
(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[3]**

4 $ABCD$ is a rectangle with length 14 cm and width 9 cm.
The point E lies on the edge CD such that $CE = 5$ cm.



NOT TO SCALE

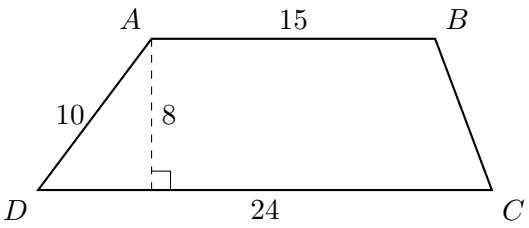
(a) Calculate the area of the shaded triangle ABE.

..... cm^2 **[2]**

(b) Calculate the perimeter of the shaded triangle ABE.

..... cm **[4]**

5 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The perpendicular height of the trapezium is 8 cm.
All measurements are in centimetres.



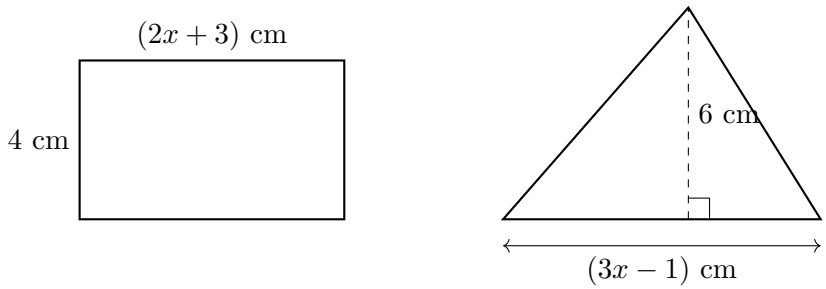
(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[4]**

6 The diagram shows a rectangle and a triangle.
The area of the rectangle is equal to the area of the triangle.



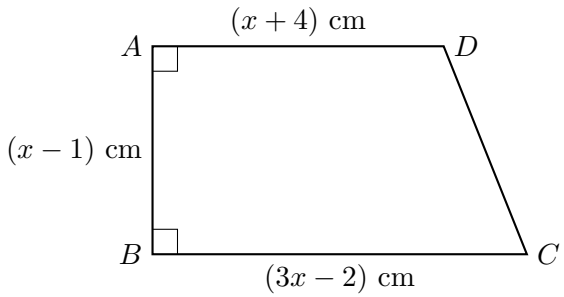
(a) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[4]**

(b) Work out the perimeter of the rectangle.

..... cm **[2]**

7 A trapezium $ABCD$ has right angles at A and B.
The side AD is parallel to BC.
$AD = (x + 4)$ cm, $BC = (3x - 2)$ cm and $AB = (x - 1)$ cm.



NOT TO SCALE

(a) The area of the trapezium is 20 cm^2.

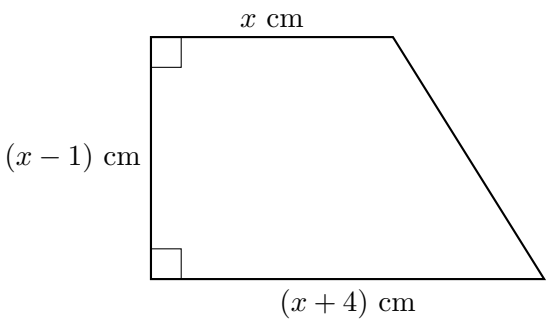
(i) Show that $2x^2 - x - 21 = 0$.

[3]

(ii) Calculate the perimeter of the trapezium.

..... cm **[5]**

8 The diagram shows a right-angled trapezium with parallel sides of length x cm and $(x + 4)$ cm.
The height of the trapezium is $(x - 1)$ cm.
The area of the trapezium is 40 cm^2.



(a) Show that $x^2 + x - 42 = 0$.

[3]

(b) Solve the equation $x^2 + x - 42 = 0$ to find the value of x.

$x =$ ..... **[2]**

(c) Calculate the perimeter of the trapezium.

..... cm **[4]**

9 A piece of wire of length 60 cm is cut into two pieces.
One piece is bent to form a square.
The other piece is bent to form a rectangle where the length is exactly twice the width.
Let the width of the rectangle be y cm.

(a) Write down an expression, in terms of y, for the side length of the square.

..... cm **[2]**

(b) The total area enclosed by the square and the rectangle is 113 cm^2.
Show that $17y^2 - 180y + 448 = 0$.

[3]

(c) Solve the equation.

$$17y^2 - 180y + 448 = 0$$

Show all your working.

$y =$ ..... , $y =$ ..... **[3]**

(d) Given that the side length of the square must be greater than the width of the rectangle, calculate the area of the square.

..... cm^2 **[2]**

1 **(a)** Calculates area of triangle: $\frac{1}{2} \times 12 \times 8 = 48$ **[M1]**

$4x = 48$

x = 12[$A1$]

(b) Uses Pythagoras to find slanted side: $\sqrt{6^2 + 8^2}$ **[M1]**

Slanted side = 10 **[A1]**

Perimeter = $12 + 10 + 10 = 32$ **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** $2(2x + 3) + 2(x - 1) = 34$ OR $2x + 3 + x - 1 = 17$ **[M1]**

$6x + 4 = 34$ OR $3x + 2 = 17$ **[M1]**

$x = 5$ **[A1]**

(b) Substitution of $x = 5$: $(2(5) + 3) \times (5 - 1)$ OR 13×4 **[M1]**

52 **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $\frac{1}{2}(10 + 16) \times 8$ **[M1]**

104 **[A1]**

(b) Base of right-angled triangle = $16 - 10 = 6$ **[M1]**

Slanted edge = $\sqrt{6^2 + 8^2} = 10$ **[M1]**

$8 + 10 + 16 + 10 = 44$ **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** Uses Area = $\frac{1}{2} \times$ base $\times$ height = $\frac{1}{2} \times 14 \times 9$ **[M1]**

63 **[A1]**

(b) Finds $AE = \sqrt{9^2 + 9^2} = \sqrt{162}$ or 12.7... **[M1]**

Finds $BE = \sqrt{5^2 + 9^2} = \sqrt{106}$ or 10.2... **[M1]**

Perimeter = $14 + \sqrt{162} + \sqrt{106}$ **[M1]**

37.0 (or 37.02...) **[A1]**

[Back to Question] *[Total Marks: 6]*

5 **(a)** $\frac{1}{2} \times (15 + 24) \times 8$ **[M1]**

156 **[A1]**

(b) Horizontal distance from D to the altitude from $A = \sqrt{10^2 - 8^2} = 6$ **[M1]**

Horizontal distance for the right-angled triangle under $B = 24 - 15 - 6 = 3$ **[M1]**

$BC = \sqrt{8^2 + 3^2}$ $(= \sqrt{73} \approx 8.544)$ **[M1]**

57.5 OR 57.54 to 57.55 **[A1]**

[Back to Question] *[Total Marks: 6]*

6 (a) Area of rectangle $= 4(2x + 3)$ OR $8x + 12$ **[M1]**

Area of triangle $= \frac{1}{2} \times 6 \times (3x - 1)$ OR $9x - 3$ **[M1]**

$8x + 12 = 9x - 3$ **[M1]**

$x = 15$ **[A1]**

(b) Length $= 2(15) + 3 = 33$ **[M1]**

Perimeter $= 2(4 + 33) = 74$ **[A1]**

[Back to Question] *[Total Marks: 6]*

7 (a) (i) Sets up equation for area: $\frac{1}{2}((x + 4) + (3x - 2))(x - 1) = 20$ **[M1]**

Simplifies expression to $\frac{1}{2}(4x + 2)(x - 1) = 20$ or $(2x + 1)(x - 1) = 20$ **[M1]**

Expands to $2x^2 - 2x + x - 1 = 20$ leading to $2x^2 - x - 21 = 0$ with no errors seen **[A1]**

(ii) Factorises $(2x - 7)(x + 3) = 0$ or uses quadratic formula to find $x = 3.5$ **[M1]**

Calculates straight sides: $AD = 7.5$, $BC = 8.5$, $AB = 2.5$ **[M1]**

Sets up Pythagoras for the slanted side CD: $\sqrt{(8.5 - 7.5)^2 + 2.5^2}$ **[M1]**

Calculates $CD = \sqrt{1^2 + 2.5^2} = \sqrt{7.25}$ (or $2.69\ldots$) **[A1]**

Perimeter $= 7.5 + 8.5 + 2.5 + 2.69\cdots = 21.2$ (or $21.19\ldots$) **[A1]**

[Back to Question] *[Total Marks: 8]*

8 (a) $\frac{1}{2}(x + x + 4)(x - 1) = 40$ **[M1]**

$\frac{1}{2}(2x + 4)(x - 1) = 40$ OR $(x + 2)(x - 1) = 40$ **[M1]**

$x^2 + 2x - x - 2 = 40 \implies x^2 + x - 42 = 0$ **[A1]**

(b) $(x + 7)(x - 6) = 0$ **[M1]**

6 **[A1]**

(c) Identifies lengths are Top $= 6$, Bottom $= 10$, Height $= 5$ **[M1]**

Horizontal difference $= 4$, so slanted side $= \sqrt{5^2 + 4^2}$ OR $\sqrt{41}$ **[M1]**

Perimeter $= 5 + 6 + 10 + \sqrt{41}$ **[M1]**

27.4 (or $27.40\ldots$) **[A1]**

[Back to Question] *[Total Marks: 9]*

9 **(a)** Perimeter of rectangle $= 6y$ OR Perimeter of square $= 60 - 6y$ **[M1]**

$\frac{60-6y}{4}$ OR $15 - 1.5y$ **[A1]**

(b) $(15 - 1.5y)^2 + y(2y) = 113$ **[M1]**

$225 - 45y + 2.25y^2 + 2y^2 = 113$ **[M1]**

$4.25y^2 - 45y + 112 = 0$ leading to $17y^2 - 180y + 448 = 0$ with no errors seen **[A1]**

(c) $\frac{-(-180) \pm \sqrt{(-180)^2 - 4(17)(448)}}{2(17)}$ **[M1]**

$\frac{180 \pm \sqrt{1936}}{34}$ OR $\frac{180 \pm 44}{34}$ **[M1]**

$y = 4$ and $y = 6.59$ (or 6.588... or $\frac{112}{17}$) **[A1]**

(d) Identifies $y = 4$ as the valid solution (since $15 - 1.5(6.59) < 6.59$) leading to side length $= 9$ **[M1]**

81 **[A1]**

[Back to Question] *[Total Marks: 10]*